Artificial Intelligence, Employment and the Future of Skills

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Abstract

This paper examines the impact of Artificial Intelligence (AI) on employment, analysing both the risks of job displacement and the opportunities for the creation of new roles and industries. Drawing on historical parallels, expert opinions and reports such as the World Economic Forum's *Future of Work 2025*, the study explores how AI is reshaping the labour market.

The research highlights the importance of continuous learning, skills development and adaptability to remain relevant in an evolving workforce, emphasising the role of human capabilities alongside technological advancement.

Over the past century, the pace of change in how societies work and live has accelerated dramatically. Each technological wave — from the Industrial Revolution to the rise of computers, the internet, cloud computing and modern operating systems — has reshaped the nature of work. While these transformations have displaced certain tasks and roles, they have also created new industries and opportunities. Those unable to adapt to these changes have often found themselves excluded from the evolving labour market.

A central question in contemporary discussions is whether artificial intelligence will replace humans in the workplace. Historically, automation has reduced demand for manual and routine tasks while simultaneously increasing demand for new forms of expertise. As with previous technological shifts, AI represents the fastest and most transformative wave to date.

As noted by chess grandmaster Garry Kasparov, artificial intelligence does not replace people; rather, people who know how to use AI replace those who do not. This perspective highlights the growing importance of AI literacy and human adaptability in modern work environments.

There is clear evidence that AI has already contributed to job displacement. In the technology sector, thousands of roles have been laid off as automated systems increasingly perform tasks previously handled by humans. Companies such as Google, Microsoft and IBM have adopted AI-driven processes, particularly in areas involving repetitive, information-based work such as data entry, routine administrative tasks and entry-level programming. Notably, approximately 25 percent of new code produced at Google is now generated by AI systems.

At the same time, AI is creating new opportunities. According to the World Economic Forum's *Future of Work 2025* report, AI is expected to stimulate job growth in sectors that have historically faced labour shortages. While the report estimates that approximately 83 million jobs may be displaced globally by 2027, it also predicts the creation of around 69 million new roles, resulting in a net loss of 14 million jobs.

These projections reinforce the global recognition that individuals must continuously develop new skills to remain relevant. Artificial intelligence is increasingly becoming a foundational technology, much like electricity, embedded across industries and daily work practices. Organisations and individuals who fail to engage in reskilling and upskilling risk long-term irrelevance in a rapidly evolving labour market.

References

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